Ziwei Gu

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Overview

Fifth-year Computer Science Ph.D. student at Harvard working on **LLM evaluation** and **human-AI interaction**. I design benchmarks and multi-turn evaluations that test how LLMs handle realistic user requests, such as ambiguous or mixed-language queries. I also build AI-powered interfaces that help people process information at scale while noticing and recovering from AI errors and steering the models.

Education

Harvard University, Cambridge, Massachusetts
Ph.D. Computer Science *Aug 2022 – May 2027 (Expected)*
Advised by Elena Glassman, Assistant Professor of Computer Science

Cornell University, Ithaca, New York
B.A. Computer Science, Magna cum laude *Aug 2017 – May 2021*
B.A. Mathematics *Aug 2017 – May 2021*

Publications

** Equal contribution*

Conference Papers

P.6 **Ziwei Gu**, Joyce Zhou, Ning-Er (Nina) Lei, Jonathan K. Kummerfeld, Mahmood Jasim, Narges Mahyar, and Elena L. Glassman. AbstractExplorer: Leveraging Structure-Mapping Theory to Enhance Comparative Close Reading at Scale. *In ACM Symposium on User Interface Software and Technology (UIST)*, 2025.

P.5 **Ziwei Gu**, Ian Arawjo, Kenneth Li, Jonathan K. Kummerfeld, and Elena L. Glassman. An AI-Resilient Text Rendering Technique for Reading and Skimming Documents. *In ACM CHI Conference on Human Factors in Computing Systems (CHI)*, 2024.

P.4 Katy Ilonka Gero, Chelse Swoopes, **Ziwei Gu**, Jonathan K. Kummerfeld, and Elena L. Glassman. Supporting Sensemaking of Large Language Model Outputs at Scale. *In ACM CHI Conference on Human Factors in Computing Systems (CHI)*, 2024. **Honorable Mention Award (top 5%)**

P.3 Jing Nathan Yan, **Ziwei Gu**, and Jeffrey M. Rzeszotarski. Tessera: Discretizing Data Analysis Workflows on a Task Level. *In ACM CHI Conference on Human Factors in Computing Systems (CHI)*, 2021.

P.2 **Ziwei Gu**[*], Jing Nathan Yan[*], and Jeffrey M. Rzeszotarski. Understanding User Sensemaking in Machine Learning Fairness Assessment Systems. *In WWW'21: The Web Conference (WWW)*, 2021.

P.1 Jing Nathan Yan, **Ziwei Gu**, Hubert Lin, and Jeffrey M. Rzeszotarski. Silva: Interactively Assessing Machine Learning Fairness Using Causality. *In ACM CHI Conference on Human Factors in Computing Systems (CHI)*, 2020.

Workshop Papers

W.2 Ajit Mallavarapu and **Ziwei Gu**. Sampling Reveals Style: Unsupervised, Training-Free Discovery of Prompt-Conditional Stylistic Axes in LLM Activations. *In the 9th Workshop on Analyzing and Interpreting Neural Networks for NLP at EMNLP (BlackboxNLP)*, 2026.

W.1 **Ziwei Gu**[*], Gauri Jain[*], Hongwen Song[*], Isak Diaz[*], Margaux Masson-Forsythe[*], and Jorge Valdes. BiomeAzuero2022: A Fine-Grained Dataset and Baselines for Tree Species Classification with Ground Images. *In the 37th AAAI Conference on Artificial Intelligence (AAAI-23) AI for Social Good Workshop*, 2023.

Extended Abstracts

EA.1 **Ziwei Gu**, Owen Raymond, Naser Al Madi, and Elena L. Glassman. Why Do Skimmers Perform Better with Grammar-Preserving Text Saliency Modulation (GP-TSM)? Evidence from an Eye Tracking Study. *In ACM CHI Conference on Human Factors in Computing Systems Late Breaking Work (CHI LBW)*, 2024.

Preprints

PP.1 **Ziwei Gu**[*], Michael S. Lee[*], Drew Rein, Yash Maurya, Vikneswaran Kumaran, Darshini Ramiah, Chenguang Wang, and Udari Madhushani Sehwag. SG-LegalClarify: a Large-scale Benchmark for Legal Reasoning and Ambiguity Recognition in Singapore Law. *Under review at ICLR 2027.*

Research Experience

Harvard University, Cambridge, Massachusetts *Aug 2022 – Present*

- Invented GP-TSM, an LLM-powered recursive sentence compression method that identifies successive levels of less important detail and renders them in lighter gray for grammar-preserving skimming; readers (N=18) scored higher on GRE reading comprehension in 12% less time than with plain text [P.5, EA.1].
- Leveraged Structure-Mapping Theory to enable comparative close reading of 1,000+ research abstracts [P.6] and sensemaking of LLM output at scale [P.4].

Cornell University, Ithaca, New York *2019 – 2021*

- Co-developed Silva, an interactive system that uses causal graphs to help users reason about sources of unfairness in machine learning datasets and models [P.1].
- Led a mixed-methods study (logs, think-alouds, interviews) of how people make sense of machine learning fairness issues with exploratory vs. recommendation-based tools [P.2].

Industry Experience

Human Frontier Collective Fellow, **Scale AI**, Remote *Dec 2025 – Present*
Mentors: Michael Lee, Udari Madhushani Sehwag

- Co-led the design and development of SG-LegalClarify, a benchmark of 1,100 lawyer-authored questions across 10 legal domains testing legal reasoning, ambiguity recognition, and selective clarification [PP.1].
- Built its multi-turn evaluation pipeline and tool-gated user simulator; evaluated 19 LLMs and found the best met only 69.8% of mandatory rubric items closed-book (77.8% with web search).
- Drove cross-functional coordination with multiple stakeholders to organize contributors across data collection and meta-evaluation.
- Designed complex, domain-specific problems and datasets to evaluate and enhance the accuracy, reasoning, and overall performance of frontier LLMs.

Research Intern, **Adobe Research**, San Jose, California *Sep 2026 – Present*
Mentors: Doga Dogan, Gromit Yeuk-Yin Chan, Eunyee Koh

- Experimented with LLM simulation strategies to estimate the visibility of unpublished drafts in AI search.
- Built a Generative Engine Optimization (GEO) Google Docs extension that lets content creators understand and test how their edits affect AI visibility as they write.
- Interviewed 9 internal GEO experts to gather insights and identify gaps in current GEO tools.

Research Intern, **Autodesk Research**, Toronto, Canada *May 2026 – Sep 2026*
Mentors: John Thompson, Justin Matejka

- Proposed Ink-to-3D-Control, a novel interaction technique that uses a VLM to turn designers' ink and speech into manipulable 3D controls that expose an agent's interpretation before any geometry changes.
- Built and evaluated InKontrol, a full-stack generative 3D-editing system (React Three Fiber, Python, CadQuery) that surfaces clarification questions and an editable build plan before generating 3D geometry.

Data Scientist Intern, **Lyft**, San Francisco, California *2020*
Mentors: Mengying Yang, Bomee Park

- Clustered millions of rider transactions into user profiles and estimated the opportunity size of Lyft Family.
- Modeled driver churn with a Markov chain; quantified incentive effects to inform engagement strategies.

Honors & Awards

Honorable Mention Award (top 5%), ACM CHI [P.4] *2024*
Magna cum laude, B.A. Computer Science, Cornell University *2021*
Statistics Faculty Award, Cornell University *2018*

Teaching Experience

Teaching Fellow, Conceptualizing, Building, and Evaluating Usable Novel Interactive Systems, Harvard CS *2026*
Head Teaching Fellow, Engineering Usable Interactive Systems, Harvard CS *2023*
Graduate Teaching Assistant, Operating Systems, Cornell CS *2021*
Teaching Assistant, Machine Learning, Cornell CS *2019 – 2020*
Teaching Assistant, Computer System Organization and Programming, Cornell CS *2019*
Teaching Assistant, Object-Oriented Programming and Data Structures, Cornell CS *2018 – 2019*

Mentoring

Ajit Mallavarapu (University of Washington) [W.2], Samhitha Goutham (Anna University), Kanishk Tantia (Harvard University), Azwad Alam (University of Southern California)

Academic Services

Reviewer
CHI (2024, 2025, 2026, 2027), UIST (2024, 2025, 2026), BlackboxNLP Workshop at EMNLP 2026, VIS 2025, Creativity & Cognition 2025

Student Volunteer
IUI 2023

Skills

Programming: Python, NumPy, Pandas, PyTorch, TensorFlow, JavaScript, React, Flask, Docker, SQL, Git

Research: ML prototyping, experimental design, agentic workflows, A/B testing, user-centered design, quantitative and qualitative analysis, data analysis, data visualization

Eval Engineering: benchmark design, rubric design, LLM-as-judge, meta-evaluation, user simulation

Outreach Activity

Treasurer, Harvard Chinese Students and Scholars Association (HCSSA) — *2023 – Present*

Fellow (Intellectual and Cultural), The Student Center at Harvard Griffin GSAS — *2023 – 2025*

Resident Advisor, Clara Dickson Hall, Cornell University — *2019 – 2021*

Project Lead, Cornell Data Science Project Team — *2018 – 2020*

Latest update: 11/2026